GAME THEORY, Spring 2026

Problem Set 1

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1 n-Bidder Auction

Consider the first/second price auction enviornment we covered in class. Instead of 2 bidders, assume there are n bidders with value $v_1, \ldots, v_n$.

(a) Extend the first price auction to this case. As usual, if more than one bidder bid the same highest price, a winning bidder is randomly drawn from them with equal probabilities. (Use this tie breaking rule also for the next question).

(b) Extend the second price auction to this case.

Answer: We write the induced normal-form game in the standard way.

(a) **First-price auction.** Let the set of players be $N = \{1, \ldots, n\}$. For each bidder $i \in N$, the strategy space is $S_i = \mathbb{R}_+$. For a bid profile $s = (s_1, \ldots, s_n) \in \prod_{i=1}^n S_i$, define the highest bid by

$$M(s) = \max_{j \in N} s_j,$$

and the set of highest bidders by

$$H(s) = \{j \in N : s_j = M(s)\}.$$

The payoff of bidder i is

$$u_i(s) = \begin{cases} \dfrac{v_i - s_i}{|H(s)|}, & i \in H(s), \\ 0, & i \notin H(s). \end{cases}$$

(b) **Second-price auction.** Again the player set is $N = \{1, \ldots, n\}$ and each bidder i chooses a bid $s_i \in S_i = \mathbb{R}_+$. Let

$$S(s) = \max\{s_j : j \in N \setminus H(s)\}$$

denote the highest bid among those who are not tied for the win. Then bidder i's payoff can be written as

$$u_i(s) = \begin{cases} v_i - S(s), & \text{if } |H(s)| = 1 \text{ and } i \in H(s), \\ \dfrac{v_i - M(s)}{|H(s)|}, & \text{if } |H(s)| \geq 2 \text{ and } i \in H(s), \\ 0, & i \notin H(s). \end{cases}$$

2 Iterated Elimination

In the following normal-form game, which strategy profiles survive iterated elimination of strictly dominated strategies?

Player 2

Player 1		L	C	R
	U	$6,8$	$2,6$	$8,2$
	M	$8,2$	$4,4$	$9,5$
	D	$8,10$	$4,6$	$6,7$

Answer: First, for Player 1, strategy U is strictly dominated by M, since

$$8 > 6, \qquad 4 > 2, \qquad 9 > 8.$$

Hence U is deleted.

After removing U, consider Player 2's payoffs in the reduced game with rows $\{M, D\}$. Now column C is strictly dominated by column R, because

$$5 > 4 \qquad \text{and} \qquad 7 > 6.$$

Hence C is deleted.

In the remaining game with rows $\{M, D\}$ and columns $\{L, R\}$, no further strict dominance exists. Therefore, the strategy profiles that survive IESDS are

$$(M, L), \quad (M, R), \quad (D, L), \quad (D, R).$$

3 Roommates

Two roommates each need to choose to clean their apartment, and each can choose an amount of time $t_i \geq 0$ to clean. If their choices are t_i and t_j, then player i's payoff is given by $(10 - t_j)t_i - t_i^2$. (This payoff function implies that the more one roommate cleans, the less valuable is cleaning for the other roommate.)

(a) What is the best response correspondence of each player i?

(b) Which choices survive one round of IESDS?

(c) Which choices survive IESDS?

Answer:

(a) Fix player j's choice $t_j \geq 0$. Player i solves

$$\max_{t_i \geq 0} u_i(t_i, t_j) = \max_{t_i \geq 0}\big((10 - t_j)t_i - t_i^2\big).$$

Since this objective is strictly concave in t_i, the best response is unique. The first-order condition is

$$\frac{\partial u_i}{\partial t_i} = 10 - t_j - 2t_i = 0,$$

so the unconstrained maximizer is $t_i = \frac{10-t_j}{2}$. Imposing $t_i \geq 0$, we obtain

$$BR_i(t_j) = \max\left\{\frac{10 - t_j}{2}, 0\right\}.$$

Equivalently,

$$BR_i(t_j) = \begin{cases} \dfrac{10 - t_j}{2}, & 0 \leq t_j \leq 10, \\ 0, & t_j \geq 10. \end{cases}$$

By symmetry, player j has the same best-response correspondence.

(b) Let $a, b \geq 0$ be two actions for player i. Then

$$u_i(a, t_j) - u_i(b, t_j) = (10 - t_j)a - a^2 - \big((10 - t_j)b - b^2\big) = (a - b)(10 - t_j - a - b).$$

If $b > 5$, then action 5 strictly dominates b, because for every $t_j \geq 0$,

$$u_i(5, t_j) - u_i(b, t_j) = (5 - b)(5 - t_j - b) > 0,$$

since $5 - b < 0$ and $5 - t_j - b < 0$ for all $t_j \geq 0$ when $b > 5$. Hence every action $b > 5$ is eliminated. On the other hand, no action in $[0, 5]$ is strictly dominated in the original game. Indeed, if $b \le 5$ and some $a > b$ were to strictly dominate b, we would need

$$(a-b)(10-t_j-a-b) > 0 \qquad \text{for all } t_j \ge 0,$$

which is impossible because for sufficiently large t_j the second factor becomes negative. If instead $a < b$, then strict dominance for all $t_j \ge 0$ would require $a+b > 10$, which cannot hold when $b \le 5$ and $a < b$. Therefore, after one round of IESDS, the surviving choices are exactly

$$[0,5].$$

(c) Let the current surviving interval be $[L,U]$. For any $a, b \in [L,U]$,

$$u_i(a,t_j) - u_i(b,t_j) = (a-b)(10-t_j-a-b).$$

If opponents are restricted to $t_j \in [L,U]$, then every action below $\frac{10-U}{2}$ is strictly dominated by $\frac{10-U}{2}$, and every action above $\frac{10-L}{2}$ is strictly dominated by $\frac{10-L}{2}$. Hence one more round of elimination maps the interval $[L,U]$ to

$$\left[\frac{10-U}{2}, \frac{10-L}{2}\right].$$

Starting from the one-round survivor set $[0,5]$, we obtain the sequence

$$[0,5] \;\to\; \left[\frac{5}{2}, 5\right] \;\to\; \left[\frac{5}{2}, \frac{15}{4}\right] \;\to\; \left[\frac{25}{8}, \frac{15}{4}\right] \;\to\; \cdots$$

These intervals converge to the unique fixed point t^* satisfying

$$t^* = \frac{10-t^*}{2} \implies t^* = \frac{10}{3}.$$

Therefore, the only choice that survives IESDS is

$$t_i = \frac{10}{3}$$

for each player.

4 Campaigning

Two candidates, 1 and 2, are running for office. Each has one of three choices in running his campaign: focus on the positive aspects of one's own platform (call this a positive campaign $[P]$), focus on the positive aspects of one's own platform while attacking one's opponent's campaign (call this a balanced campaign $[B]$), or finally focus only on attacking one's opponent (call this a negative campaign [N]). All a candidate cares about is the probability of winning, so assume that if a candidate expects to win with probability $\pi \in [0,1]$, then his payoff is π. The probability that a candidate wins depends on his choice of campaign and his opponent's choice. The probabilities of winning are given as follows:

If both choose the same campaign then each wins with probability 0.5.
If candidate i uses a positive campaign while $j \neq i$ uses a balanced one then i loses for sure.
If candidate i uses a positive campaign while $j \neq i$ uses a negative one then i wins with probability 0.3.
If candidate i uses a negative campaign while $j \neq i$ uses a balanced one then i wins with probability 0.6.

(a) Model this story as a normal-form game. (It suffices to be specific about the payoff function of one player and to explain how the other player's payoff function is different and why.)
(b) Write down the game in matrix form.
(c) What happens at each stage of elimination of strictly dominated strategies? Will this procedure lead to a clear prediction?

Answer:

(a) Let the set of players be $N = \{1, 2\}$, and each player's strategy set be

$$S_i = \{P, B, N\}, \qquad i \in \{1, 2\}.$$

For player 1, the payoff function $u_1 : S_1 \times S_2 \to [0,1]$ is given by player 1's probability of winning:

$$u_1(P,P) = u_1(B,B) = u_1(N,N) = 0.5,$$

$$u_1(P,B) = 0, \qquad u_1(B,P) = 1,$$

$$u_1(P,N) = 0.3, \qquad u_1(N,P) = 0.7,$$

$$u_1(N,B) = 0.6, \qquad u_1(B,N) = 0.4.$$

Since the game is symmetric and one candidate's probability of winning is the other candidate's probability of losing, player 2's payoff function is given by

$$u_2(s_1, s_2) = 1 - u_1(s_1, s_2).$$

(b) The game in matrix form is therefore

Candidate 2

Candidate 1		P	B	N
	P	$0.5, 0.5$	$0, 1$	$0.3, 0.7$
	B	$1, 0$	$0.5, 0.5$	$0.4, 0.6$
	N	$0.7, 0.3$	$0.6, 0.4$	$0.5, 0.5$

(c) First, for each player, strategy P is strictly dominated by N, so P is eliminated for both players. After deleting P, the reduced game has strategies $\{B, N\}$ for each player. In this reduced game, for each player, B is strictly dominated by N. Therefore B is eliminated as well, and the only surviving strategy for each player is N. Hence the unique strategy profile that survives IESDS is

$$(N, N).$$

So in this game, IESDS does lead to a clear prediction: both candidates run negative campaigns.

5 Beauty Contest Best Responses

Consider a game with n players, so that $N = \{1, \ldots, n\}$. Each player has to choose an integer between 0 and 20, so that $S_i = \{0, 1, 2, \ldots, 19, 20\}$. The winners are the players who choose an integer number that is closest to $\frac{3}{4}$ of the average.

Because more than one person can win, define th set of winners $W \subset N$ as those players who are closest to $\frac{3}{4}$ of the average, and the rest are all losers. The set of winners is defined as

$$W = \left\{ i \in N : \arg\min_{i \in N} \left| s_i - \frac{3}{4}\frac{1}{n}\sum_{j=1}^{n} s_j \right| \right\}$$

To introduce payoffs, each player pays 1 to play the game, and winners split the pot equally among themselves. This implies that if there are $m \geq 1$ winners, each gets a payoff of $\frac{N-m}{m}$ (his share of the pot net of his own contribution) while losers get -1 (they lose their contribution).

(a) Show that if player i believes that everyone else is choosing 20 then 19 is not the only best response for any number of players n.

(b) Show that the set of best-response strategies to everyone else choosing the number 20 depends on the number of players n.

Answer:

(a) Suppose every player other than i chooses 20, and let player i choose $x \in \{0, 1, \ldots, 20\}$.

Then the average is

$$\bar{s}(x) = \frac{20(n-1)+x}{n},$$

so the target number is

$$T(x) = \frac{3}{4}\bar{s}(x) = \frac{3}{4} \cdot \frac{20(n-1)+x}{n} = 15 - \frac{15}{n} + \frac{3x}{4n}.$$

Consider first $x = 19$. Then

$$T(19) = 15 - \frac{3}{4n},$$

so player i's distance from the target is

$$|19 - T(19)| = 4 + \frac{3}{4n},$$

while each player choosing 20 is at distance

$$|20 - T(19)| = 5 + \frac{3}{4n}.$$

Hence 19 makes player i the unique winner.

Now consider $x = 18$. Then

$$T(18) = 15 - \frac{3}{2n},$$

and therefore

$$|18 - T(18)| = 3 + \frac{3}{2n}, \qquad |20 - T(18)| = 5 + \frac{3}{2n}.$$

So 18 also makes player i the unique winner.

If player i is the unique winner, his payoff is $n-1$. Thus both 18 and 19 yield payoff $n-1$, so 19 is *not* the only best response for any n.

(b) Again let player i choose $x \in \{0, 1, \dots, 20\}$ while every other player chooses 20. If $x = 20$, then everyone ties as a winner, so player i's payoff is

$$\frac{n-n}{n} = 0.$$

If $x < 20$, then the other $n-1$ players all choose the same number, so either player i is the unique winner or else he loses. Hence for $x < 20$,

$$u_i(x, 20, \dots, 20) = \begin{cases} n-1, & \text{if } |x - T(x)| < |20 - T(x)|, \\ -1, & \text{if } |x - T(x)| > |20 - T(x)|. \end{cases}$$

Therefore, the best responses are exactly those $x < 20$ for which player i is strictly closer to the target than 20 is.

We solve

$$|x - T(x)| < |20 - T(x)|.$$

Since $x < 20$, we have $x - 20 < 0$, so after squaring both sides and factoring,

$$(x-T(x))^2 < (20-T(x))^2 \iff (x-20)(x+20-2T(x)) < 0 \iff x+20-2T(x) > 0.$$

Substituting for $T(x)$ gives

$$x + 20 - 2\left(15 - \frac{15}{n} + \frac{3x}{4n}\right) > 0,$$

which simplifies to

$$x\left(1 - \frac{3}{2n}\right) > 10 - \frac{30}{n} \iff x(2n-3) > 20(n-3).$$

Thus the set of best responses is

$$BR_i(20, \ldots, 20) = \left\{x \in \{0, 1, \ldots, 19\} : x > \frac{20(n-3)}{2n-3}\right\}.$$

This set depends on n. For example,

$$BR_i(20) = \{0, 1, \ldots, 19\} \quad \text{when } n = 2,$$

while

$$BR_i(20, 20, 20) = \{5, 6, \ldots, 19\} \quad \text{when } n = 4.$$

Hence the set of best-response strategies to everyone else choosing 20 does indeed depend on the number of players n.

6 IESDS and BR in Infinite Games

Consider the following two-player game. Each player announces a nonnegative real number. The payoffs are

$$v_i(x_i, x_j) = \begin{cases} 2, & \text{if } x_i = 0 \text{ and } x_j = 1, \\ \arctan x_i, & \text{if otherwise.} \end{cases}$$

(a) Argue that every positive announcement is strictly dominated.

(b) Argue that announcement 0 is not strictly dominated.

(c) From the above two questions, we know only 0 survives IESDS for both players. Are they mutual best responses?

Answer:

(a) Fix any announcement $x_i > 0$. We claim that x_i is strictly dominated by any larger announcement $y_i > x_i$. Indeed, for every announcement $x_j \geq 0$ of player j, if player i chooses $x_i > 0$, then the special case in the payoff definition never applies, so

$$v_i(x_i, x_j) = \arctan x_i.$$

Likewise,

$$v_i(y_i, x_j) = \arctan y_i.$$

Since $\arctan(\cdot)$ is strictly increasing on $\mathbb{R}_+$ and $y_i > x_i$, we have

$$\arctan y_i > \arctan x_i.$$

Therefore,

$$v_i(y_i, x_j) > v_i(x_i, x_j) \qquad \text{for all } x_j \geq 0,$$

so every positive announcement is strictly dominated.

(b) Let $y_i \geq 0$ be any alternative announcement. If $y_i = 0$, then y_i clearly does not strictly dominate 0, since the two strategies are identical.

If $y_i > 0$, consider the opponent announcement $x_j = 1$. Then

$$v_i(0, 1) = 2,$$

while

$$v_i(y_i, 1) = \arctan y_i < \frac{\pi}{2} < 2.$$

Hence y_i cannot strictly dominate 0. Since no strategy $y_i \geq 0$ strictly dominates 0, announcement 0 is not strictly dominated.

(c) From parts (a) and (b), the only strategy that survives IESDS is 0 for each player. However, these are *not* mutual best responses.

To see this, fix $x_j = 0$. Then for player i,

$$v_i(x_i, 0) = \arctan x_i \qquad \text{for all } x_i \geq 0,$$

because the exceptional payoff 2 occurs only when $(x_i, x_j) = (0, 1)$. Since $\arctan x_i$ is strictly increasing in x_i, we have

$$\arctan x_i > \arctan 0 = 0 \qquad \text{for every } x_i > 0.$$

So $x_i = 0$ is not a best response to $x_j = 0$.
In fact, there is no best response to $x_j = 0$ at all: the payoff $\arctan x_i$ can be made arbitrarily close to $\pi/2$ by choosing larger and larger x_i, but it never reaches $\pi/2$ for any finite x_i. Thus the best-response set to 0 is empty.
Therefore, although IESDS leaves only the strategy profile $(0,0)$, that profile does not consist of mutual best responses.

7 n-Firm Cournot Competition

Consider the n-firm Cournot competition. The demand curve is still

$$D(Q) = \max\{100 - Q, 0\}.$$

If each firm i supplies q_i, the total supply is $\sum_{i=1}^{n} q_i$. Suppose each firm's marginal cost is 10.

(a) Write down its normal form game.
(b) For each firm, what are the strategies that survive IESDS?

Answer:

(a) Let the set of firms be $N = \{1, \ldots, n\}$. Firm i chooses an output level $q_i \in S_i = \mathbb{R}_+$. Since each firm has constant marginal cost 10, firm i's profit is

$$v_i(q_1, \ldots, q_n) = q_i\big(D(Q) - 10\big) = q_i \left(\max\left\{ 100 - \sum_{k=1}^{n} q_k, 0 \right\} - 10 \right).$$

Equivalently,

$$v_i(q) = \begin{cases} q_i\left(90 - \sum_{k=1}^{n} q_k\right), & \text{if } \sum_{k=1}^{n} q_k \le 100, \\ -10 q_i, & \text{if } \sum_{k=1}^{n} q_k > 100. \end{cases}$$

Therefore the normal-form game is

$$G = \big(N, (S_i)_{i \in N}, (v_i)_{i \in N}\big)$$

with $N = \{1, \ldots, n\}$, $S_i = \mathbb{R}_+$ for every i, and payoff functions v_i as above.

(b) We first derive firm i's best response to the total output of the other firms. Let

$$x = \sum_{j \ne i} q_j.$$

Then firm i chooses $q_i \geq 0$ to maximize

$$q_i\big(\max\{100 - x - q_i, 0\} - 10\big).$$

If $x \geq 90$, every positive q_i yields negative profit, so the best response is $q_i = 0$. If $x < 90$, then the relevant objective is

$$q_i(90 - x - q_i),$$

whose unique maximizer is obtained from the first-order condition

$$90 - x - 2q_i = 0.$$

Thus

$$BR_i(x) = \max\left\{\frac{90 - x}{2}, 0\right\}.$$

We now perform IESDS.

First, every quantity $q_i > 90$ is strictly dominated by 0, because for every profile of the other firms,

$$v_i(q_i, q_{-i}) < 0 = v_i(0, q_{-i}).$$

Hence after the first round each firm's surviving strategies lie in

$$[0, 90].$$

Next, in this reduced game every quantity $q_i > 45$ is strictly dominated by 45. Indeed, let $x = \sum_{j \neq i} q_j \geq 0$ and take any $q_i > 45$. If $x + q_i \leq 100$, then both profits are computed in the positive-price region, and

$$v_i(45, x) - v_i(q_i, x) = (45 - q_i)(45 - x - q_i) > 0,$$

because $45 - q_i < 0$ and $45 - x - q_i < 0$. If $x + q_i > 100$, then $v_i(q_i, x) = -10q_i < -450$. On the other hand, if $x \leq 55$, then

$$v_i(45, x) = 45(45 - x) \geq -450,$$

while if $x > 55$, then also $x + 45 > 100$, so

$$v_i(45, x) = -450.$$

In either case $v_i(45, x) > v_i(q_i, x)$. Hence every quantity above 45 is eliminated. Therefore, after two rounds, each firm's surviving strategies are contained in

$$[0, 45].$$

At this point the answer depends on n.

If $n = 2$, let the current surviving interval be $[L, U] \subseteq [0, 45]$. Since the opponent's output lies in $[L, U]$, firm i's best responses lie in

$$\left[\frac{90-U}{2}, \frac{90-L}{2}\right].$$

Hence one more round of elimination maps $[L, U]$ to

$$\left[\frac{90-U}{2}, \frac{90-L}{2}\right].$$

Starting from $[0, 45]$, we obtain

$$[0, 45] \to \left[\frac{45}{2}, 45\right] \to \left[\frac{45}{2}, \frac{135}{4}\right] \to \cdots$$

and these intervals converge to the unique fixed point q^* satisfying

$$q^* = \frac{90-q^*}{2}.$$

Solving gives

$$q^* = 30.$$

So when $n = 2$, the only strategy that survives IESDS is

$$q_i = 30.$$

If $n \geq 3$, no further elimination is possible after reaching $[0, 45]$. To see this, fix any $q \in [0, 45]$. Then

$$x = 90 - 2q \in [0, 90].$$

Because there are at least two rival firms, we can choose their outputs in $[0, 45]$ so that their total output is exactly x (for instance, let one rival produce $\min\{45, x\}$, another produce the remainder, and any further rivals produce 0). Against such a profile, firm i's best response is

$$BR_i(x) = \frac{90-x}{2} = q.$$

Hence every $q \in [0, 45]$ is a best response to some surviving profile of the other firms, so no such strategy can be strictly dominated. Therefore, when $n \geq 3$, the strategies that survive IESDS are exactly

$$[0, 45].$$